

# Cervical Cancer Screening — Clinical Reference

Persistent hrHPV (16/18) → >90% at transformation zone · TeachIM.org (Dec 2022, optimized Jun 2026) · CC BY-NC 4.0 · educational use only — clinician-review before teaching

## 1 · Natural history

- Persistent high-risk HPV (chiefly 16, 18) causes nearly all cervical cancer; ~90% of infections clear within 2 y, persistent minority → precancer over years.
- >90% arise at the **transformation zone** (squamocolumnar junction) — sample it. Ectocervix (squamous) → SCC ~75%; endocervix (columnar) → adenocarcinoma ~25%.
- Persistence risk ↑: older age, oncogenic subtype, immunosuppression, smoking, other STIs, ?OCP.

## 2 · Average-risk screening

| Age       | Options                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------|
| 21–29     | Cytology q3y (no HPV testing <30)                                                                |
| 30–65     | Primary hrHPV q5y (USPSTF 2025 / ACS preferred) — or co-test q5y, or cytology q3y                |
| Stop      | >65 if no CIN2+ x25y + adequate prior screen (3 neg cyt/10y or 2 neg HPV/10y, most recent <3/5y) |
| Post-hyst | Stop after total hysterectomy (cervix out) for benign cause + no CIN2+ x25y                      |

- HPV vaccination does NOT change the schedule. Positive primary HPV → reflex cytology, same specimen.

## 3 · High-risk (immunosuppressed)

- HIV, transplant, SLE, IBD/RA on immunosuppression, chronic steroids.
- **HIV:** <30 cytology only (baseline → annual x3 → q3y); ≥30 cytology or co-test q3y; **continue for life** (no stop at 65). Primary HPV-with-reflex NOT recommended.
- Non-HIV immunosuppressed → follow CDC HIV protocol. IBD/RA off immunosuppression or T1DM → general-population guideline.
- Lower threshold for colposcopy: ASC-US/HPV+, or LSIL/ASC-H/AGC/HSIL regardless of HPV.

## 4 · Manage abnormal results (ASCCP app)

| Immediate CIN3+ risk | Action                                                              |
|----------------------|---------------------------------------------------------------------|
| <4%                  | Surveillance by 5-y risk: >0.55% → 1y; 0.15–0.55% → 3y; <0.15% → 5y |
| 4–24%                | Colposcopy                                                          |
| 25–59%               | Expedited tx or colposcopy (per future-pregnancy desire)            |
| 60–100%              | Expedited treatment (excision > ablation)                           |

- **Cytology (↑ severity):** NILM → ASC-US → LSIL → ASC-H → AGC → HSIL. **Histology:** No CIN → CIN1 → CIN2-3 → AIS.
- Excision (LEEP/cone) preferred over ablation for histologic HSIL & AIS (yields tissue).
- **Hx CIN2+:** HPV-based at 6 mo → 3 annual → q3y for ≥25 y (even >65).

## 5 · 2025 update (framework unchanged)

- **USPSTF 2025 (Grade A):** primary hrHPV q5y preferred for 30–65; cytology q3y still an option.
- First-time endorsement of FDA-approved **self-collected** hrHPV (CIN2+ Sn ~0.86) — access expansion.
- **ACS 2025:** start at 25 with primary hrHPV; self-collection endorsed.

Sources: USPSTF 2018 & final 2025; ACS 2020/2025 (CA Cancer J Clin); ASCCP 2019 risk-based guidelines + Oct 2025 self-collection advisory; CDC/NIH/IDSA HIV OI guidelines. Adapted from TeachIM.org under CC BY-NC 4.0.